

Anish Patnaik

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EDUCATION

University of California – San Diego, San Diego, CA

Sep 2025 - Dec 2026

Master of Science – Data Science (GPA: 4.0/4.0)

Relevant Coursework: Efficient AI, Web Mining and Recommender Systems, Probability and Statistics

Manipal Institute of Technology, Manipal, India

Jul 2018 - May 2022

Bachelor of Engineering - Electronics and Communication Engineering (GPA: 9.39/10)

Minor Specialization: Data Science

Relevant Coursework: Calculus, Advanced Data Science, Machine Learning, Computer Vision

SKILLS

- **Programming Languages:** Python, C++, SQL, R
- **Deep Learning & GenAI:** PyTorch, TensorFlow, Transformers, LLMs, RAG, NLP, Computer Vision, Prompt Engineering
- **Data Science & Machine Learning:** Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, NLTK, Feature Engineering, A/B Testing, Hypothesis Testing, Power BI
- **Cloud & MLOps:** Microsoft Azure (Functions, Synapse Analytics, ML Studio, Blob Storage, Azure SQL), Github, Databases, Docker, CI/CD, ML System Design, Serverless Architecture

PROFESSIONAL EXPERIENCE

Optum – UnitedHealth Group, Hyderabad, India

Jul 2022 - Aug 2025

Associate Data Scientist

- **Engineered** an end-to-end DiT & OpenAI LLM pipeline to classify appeal intents, achieving **97.5% accuracy** and slashing processing time by **15%** (saving **~\$1.4M annually**).
- **Built and deployed** a real-time access risk review model using **Azure Functions**, serving risk scores for **100k+ accesses** with **under 50ms latency** via a serverless architecture.
- **Prototyped** a **Random Forest** access recommendation model (**96.1% AUC**) by designing a scalable **Jaccard similarity** engine, synthesizing features from **peer access overlap** for **400k+ employees** and exposing via **internal APIs** for organization-wide utilization.
- **Reduced** new hire attrition from **18% to 11%** (saving **~\$0.5M annually**) by implementing **EDA workflows** on hiring data across **29 business lines** to drive strategic retention initiatives.

ConceptWaves Software Solutions, Hyderabad, India

Jan 2022 - May 2022

Machine Learning Intern

- **Developed** an XGBoost model to predict MBA admission chances with **92% accuracy** by engineering features from academic credentials and extracurricular data
- **Optimized** model performance by **15%** through rigorous feature importance analysis and packaged the workflow into **reproducible Jupyter notebooks** for client integration.

PROJECTS

UCSD SMASH x NSF HDR ML Challenge

Jan 2026

- **Secured 2nd Place** by developing a robust machine learning model to **predict coastal flooding levels**, successfully generalizing across unseen geographic locations (**Out-of-Distribution data**).
- **Engineered** a reproducible inference pipeline that processed environmental sensor data to handle **data shift**, achieving top-tier performance on the **Codabench** leaderboard.

MyTorch: Custom Deep Learning library built from scratch

Dec 2025 - Jan 2026

- **Reverse-engineered** PyTorch mechanics to build a custom DL library from scratch, implementing a full **forward/backward propagation engine** and custom **SGD optimizer**.
- **Architected** modular components including Linear layers, Batch Normalization, and Activations (ReLU, Sigmoid, Tanh) to successfully construct and train Multilayer Perceptrons (MLPs).

Neonatal seizure detection using EEG Signals with Deep learning methods

Jan 2024 - Nov 2024

- **Achieved 97.49% accuracy** in seizure prediction across **12,000+ EEG samples** by designing a novel **CNN–Attention ensemble** architecture.
- **Outperformed** baseline **ResNet50** and **1D/2D CNN** models through extensive benchmarking and robust **preprocessing** (ICA, windowing, and normalization).